

Deep models of protein evolution in time generate realistic evolutionary trajectories and functional proteins

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Vast diversity of life

8.7 million $\pm$ 1.3 million eukaryotic species (Mora et al. 2011)

- ~6.5 million on land
- ~2.2 million in the ocean

Many more microbes (bacteria, archaea, viruses)

- 1 trillion microbial lineages (?)

Numerous individuals per species

Evolution has done massive-scale natural “experiments” with:

- Point mutations
- Insertions and deletions
- Genome rearrangements
- Different combinations of existing and new genes
- MANY replicates

Deep Models of Protein Sequences

There are many protein language models (pLMs) that describe the distribution $p(x)$ of protein sequences x that exist today.

- **Masked:** ESM, TAPE, ProtTrans, MSA Transformer, CARP, ...
- **Autoregressive:** ProGen, Tranception, ...

Superb at **modeling interactions** between sites.

Particularly effective in structure prediction, variant effect prediction, and protein design.

Implicitly treat sequences as being i.i.d., leading to **phylogenetic biases**.

- Typically addressed by evolutionary de-duplication using prior clustering
- Discards local sequence variation important for specialized function

Cannot inherently describe **how proteins evolve over time**.

- Lacks evolutionary reasoning

Part I

What is an evolutionary model?

The central question

Given:

- a starting protein sequence (x);
- an evolutionary distance (t);

what sequences (y) could plausibly evolve?

$$p(y \mid x, t)$$

This conditional distribution is an **evolutionary model**.

What does “time” mean here?

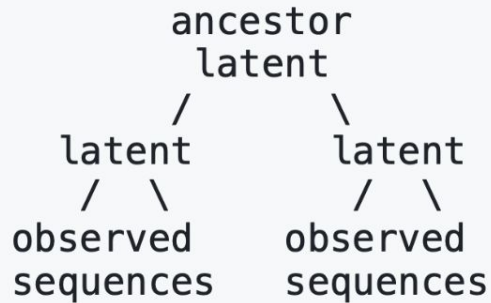
Branch length (t) is usually **evolutionary distance**, not calendar time.

- Often normalized as expected substitutions per site
- Small (t): descendants should resemble the starting sequence
- Large (t): more substitutions—and possibly indels—accumulate
- Predictions should change smoothly as (t) changes

small t
 $x \longrightarrow$ similar y

large t
 $x \longrightarrow$ divergent y

We only observe the leaves



Present-day proteins are observed.

Ancestral proteins and the transitions between them are usually hidden.

Evolutionary modeling is therefore a **latent-variable problem**.

A phylogenetic model

For a tree (T):

1. Draw a root sequence:

$$x_{\text{root}} \sim \pi_{\text{root}}$$

2. Along each edge $((u,v,t))$, evolve:

$$x_v \sim p(x_v \mid x_u, t)$$

3. Repeat until all leaves have sequences.

The tree turns a local transition rule into a probability model for an entire protein family.

From protein families to training data

Typical inputs include:

- many protein families $(T_1, \dots, T_m)_i$;
- observed leaf sequences $(D_1, \dots, D_m)_i$;
- sequences organized into multiple sequence alignments;
- estimated phylogenetic trees.

Scales discussed in the paper:

$$m \approx 10^4, \quad n \approx 10^3, \quad l \approx 200$$

That corresponds to roughly (10^9) amino-acid tokens.

Classical model: independent sites

Aligned sites are assumed to evolve independently:

$$p(y \mid x, t) = \prod_{i=1}^l p(y_i \mid x_i, t)$$

Why this assumption is useful

- Efficient likelihood computation
- Interpretable parameters
- Tractable ancestral-state inference

What it removes

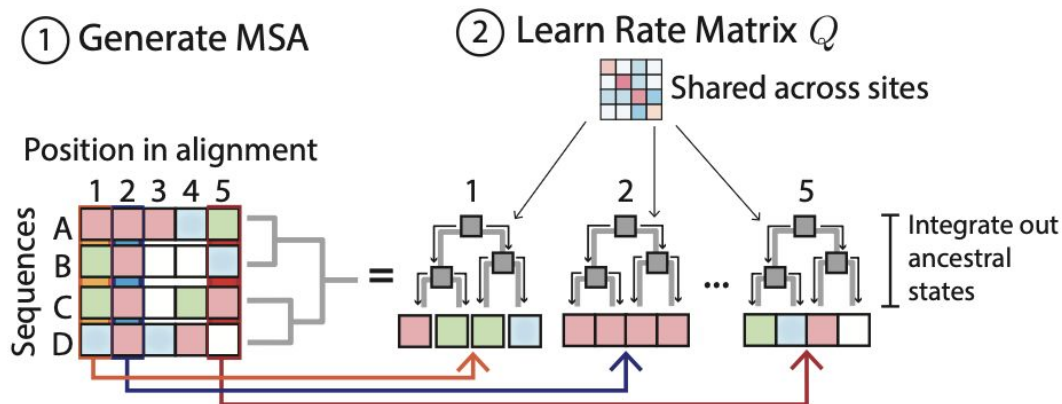
- Residue interactions
- Context-dependent substitutions
- Explicit insertion and deletion processes

Continuous-time Markov chains

A (20×20) rate matrix (Q) describes instantaneous amino-acid changes.

Transition probabilities after evolutionary distance (t) are:

$$P(t) = \exp(tQ)$$



Sites evolve independently under shared rate matrix and tree

$\mathbb{P}(\text{sequences}) = \text{a product of marginal probabilities}$

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For a WAG-like independent-site model:

$$p(y \mid x, t) = \prod_{i=1}^l \exp(tQ)[x_i, y_i]$$

Alignment gaps are generally treated as missing data rather than modeled evolutionary events.

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WAG versus LG

LG adds family-specific site rates (α_i):

$$p(y \mid x, t, c) = \prod_i \exp(\alpha_i t Q)[x_i, y_i]$$

The independent-site assumption

Translating a protein one position at a time is like translating a sentence one word at a time—without looking at the other words.

Background: epistasis

Protein epistasis is the phenomenon where the functional or fitness effect of a mutation in a protein sequence depends on the presence of other mutations.

- A mutation may be neutral in one background and harmful in another.
- Two mutations can have non-additive effects.
- Compensatory mutations can restore structure or function.
- Higher-order epistasis can involve many interacting sites.

Why full inference becomes difficult

With independent sites, dynamic programming integrates over hidden ancestral residues:

$$\mathcal{O}(mnl s^2), \quad s = 20$$

Once sites interact, the state is an entire sequence:

$$s^l = 20^l$$

Exact marginalization over all possible ancestral proteins becomes intractable.

Expressiveness breaks the classical computational shortcut.

Cherries: learning from close leaf pairs

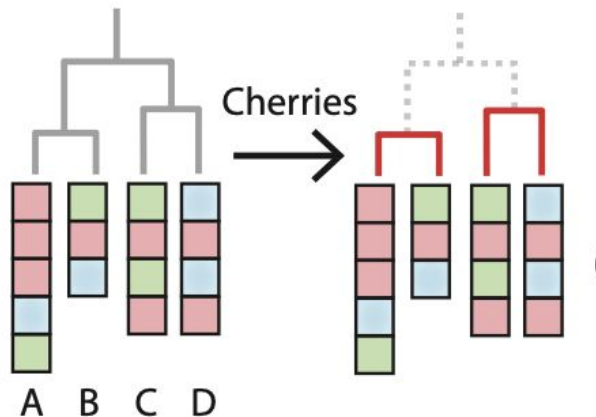
A **cherry** is a pair of leaves sharing a recent ancestor.

Instead of optimizing the full tree likelihood, use a composite likelihood:

$$\ell_{\text{comp}} = \sum_{i=1}^m \sum_{j=1}^{c_i} \log p_{\text{phylo}} (D_i[u_{ij}] \mid D_i[v_{ij}], t_{ij})$$

For a reversible model, only the pair and their evolutionary distance are needed.

This converts a difficult latent-tree problem into many tractable conditional transition problems.



Part II

From classical models to PEINT

The modeling gap

Classical CTMCs

- Explicit evolutionary time & inference
- Independent aligned sites
- No explicit indel generation

Protein language models

- Strong sequence context, structural and functional signals
- Usually treat sequences as independent samples
- No explicit branch-length conditioning

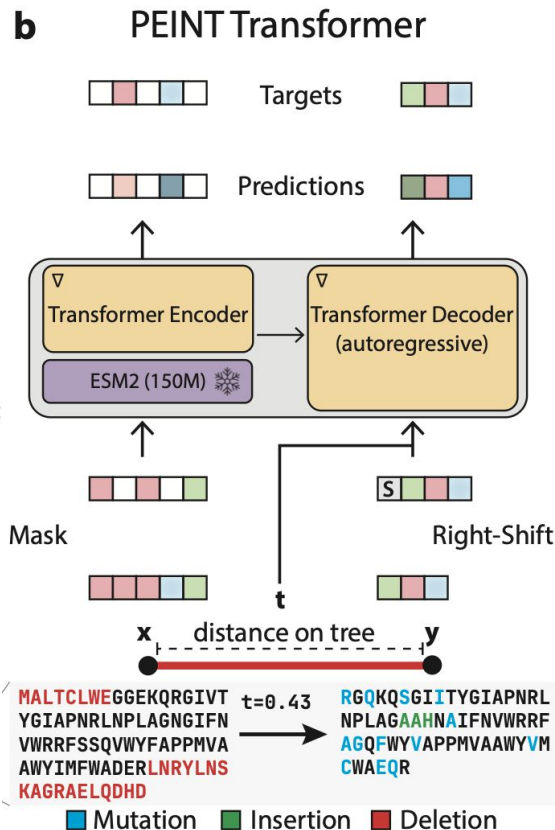
We need a model that knows both **context and **time**.**

PEINT: Protein Evolution IN Time

- **Input:** source sequence (x)
- **Condition:** evolutionary distance (t)
- **Output:** evolved sequence (y)
- **Architecture:** encoder–decoder transformer
- **Training signal:** transitions extracted from phylogenetic

PEINT treats evolution like translation:

translate a protein from its past version to a future version, while conditioning on the distance between them on the tree.



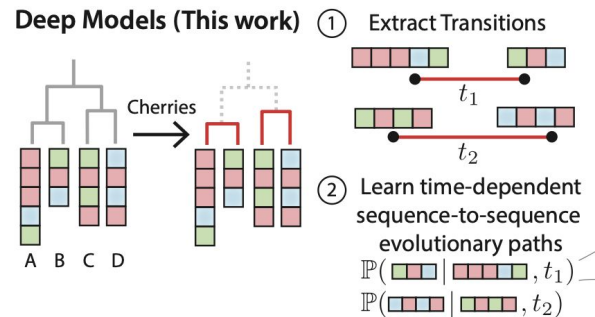
How are training examples constructed?

Training examples are sequence-transition triples:

(x, y, t)

- Infer trees from protein-family MSAs.
- Extract related sequence pairs.
- Associate each pair with its phylogenetic distance.
- Train the model to generate one sequence from the other.

$$\mathcal{L} = \sum_T \sum_{(x,y,t) \in T} \log p_{\theta}(y \mid x, t)$$



Alignment-free evolution

PEINT trains on native, ungapped sequences.

- Classical models operate on aligned columns.
- PEINT removes alignment gaps before modeling transitions.
- The generated sequence may be longer or shorter than the input.
- Insertions and deletions become part of the generative process.
- The model is less dependent on potentially incorrect alignments.

PEINT architecture

Time-conditioned encoder–decoder transformer

- Frozen ESM2-150M representations
- Five additional encoder layers
- Five decoder layers
- Approximately 57 million trainable parameters
- Continuous embedding of evolutionary time

Training objective

PEINT combines two objectives.

Source-sequence encoder

Masked-language-modeling objective on (x)

Evolved-sequence decoder

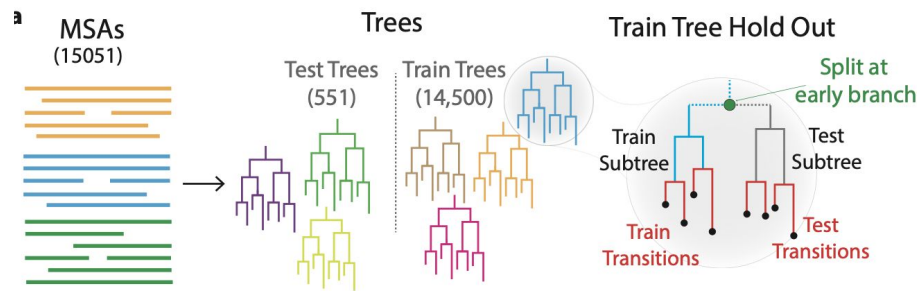
$$p(y \mid x, t) = \prod_{i=1}^L p(y_i \mid y_{<i}, x, t)$$

Each generated residue can depend on:

- the complete source sequence;
- evolutionary time;
- all previously generated residues.

Dataset and evaluation splits

- TrRosetta-derived dataset
- Approximately 15,000 protein-family MSAs
- Trees inferred from the MSAs
- Approximately 14,500 training families
- 553 fully held-out test families
- Millions of extracted evolutionary transitions



Two generalization settings:

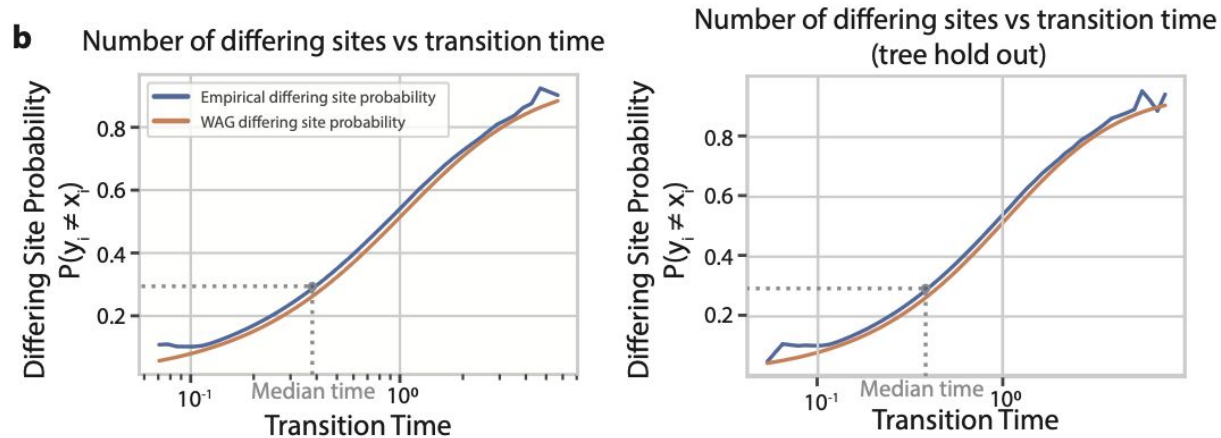
1. **In-family holdout:** evaluate on a separate subtree.
2. **Held-out families:** evaluate on entirely unseen protein families.

Dataset and evaluation splits

Empirical and theoretical measures of sequence divergences as a function of time.

The median transition has 30% sites that differ between x and y .

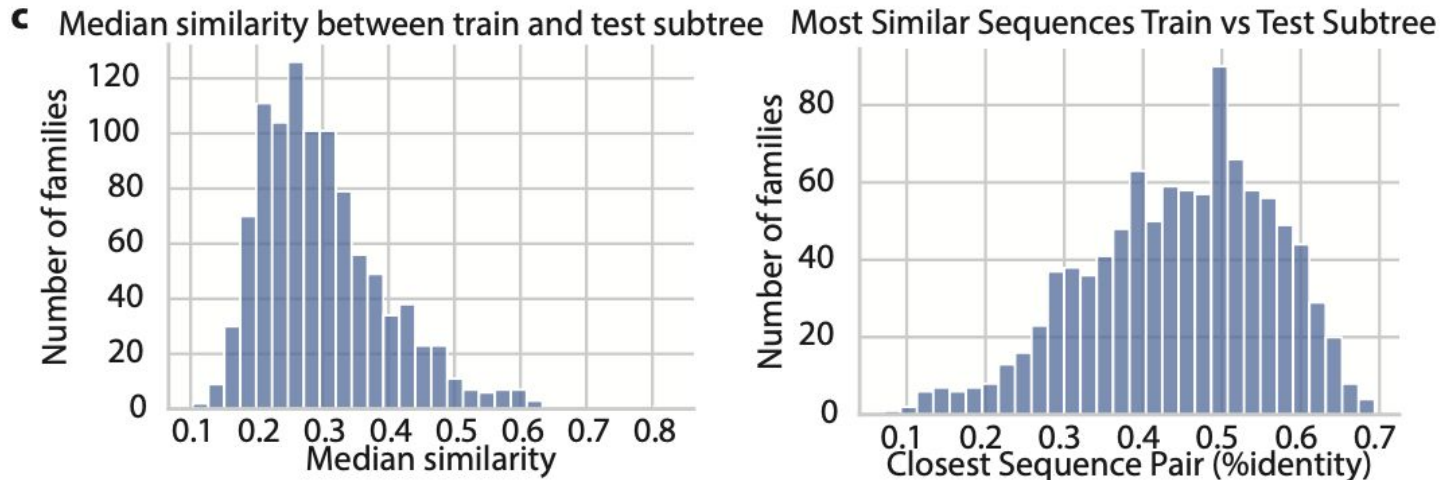
Left is from the training subtrees, right is from the held-out subtrees, showing similar statistics.



Dataset and evaluation splits

Median all-vs-all pairwise percent identity (left) and closest pair of sequences (right) between held-out and training subtrees.

The two trees show minimal overlap in sequence space

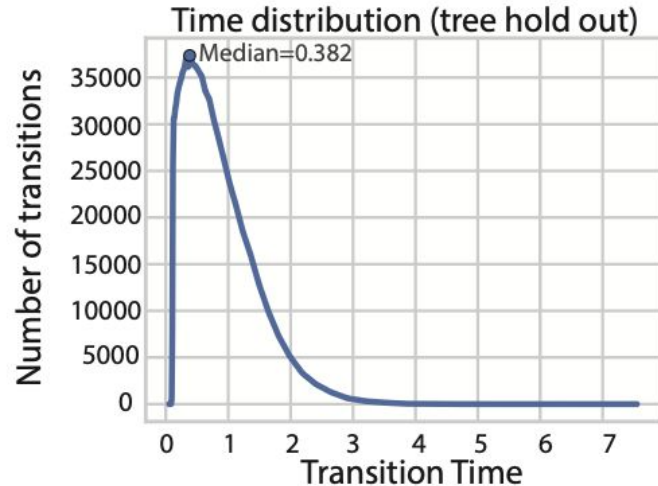
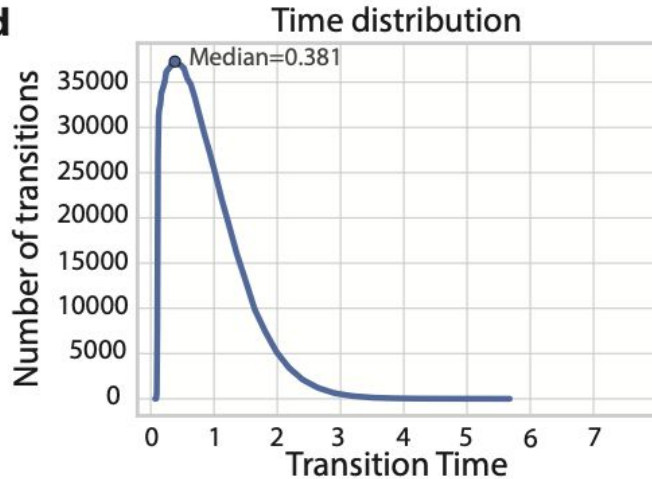


Dataset and evaluation splits

Distribution of transition times from the cherries in the training and held-out subtree sets.

Training and held-out subtrees show similar distributions of cherry branch lengths.

d



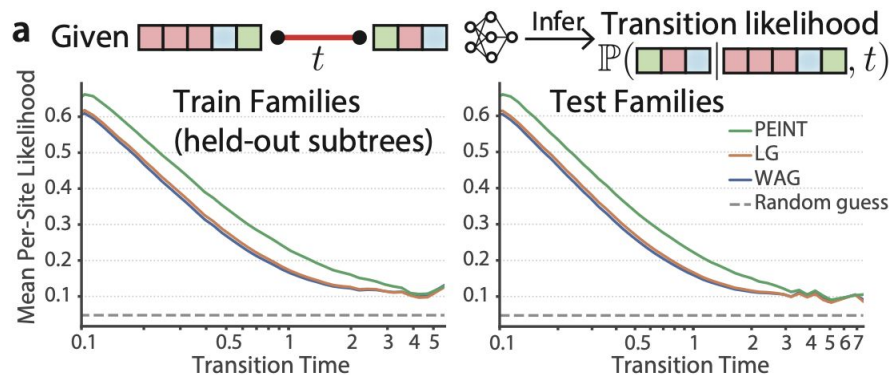
Part III

Does PEINT model realistic evolution?

Evaluating evolutionary transitions along single branches

Given a held-out transition (x, y, t) , how much probability does the model assign to the observed (y) ?

- PEINT gives higher per-site likelihood across transition times.



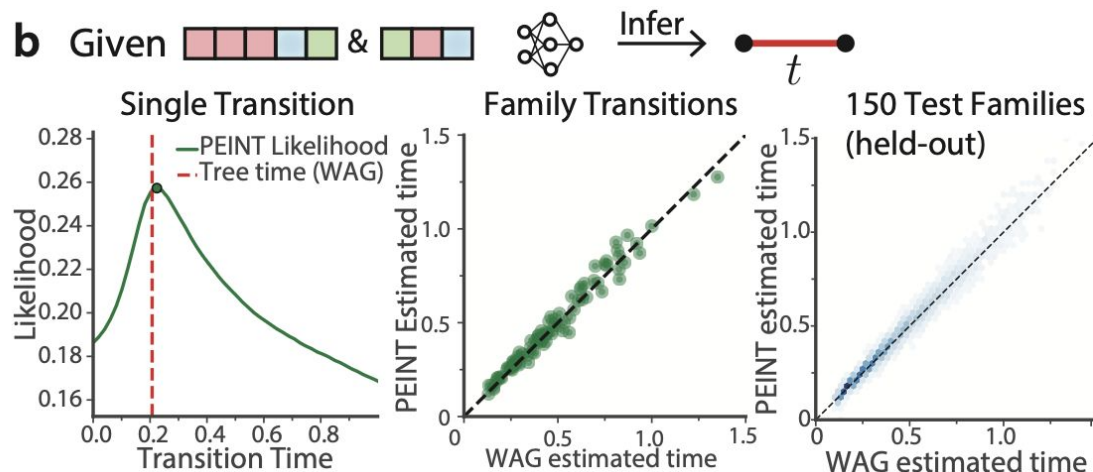
PEINT can infer branch lengths

Can PEINT recover evolutionary distance from two sequences?

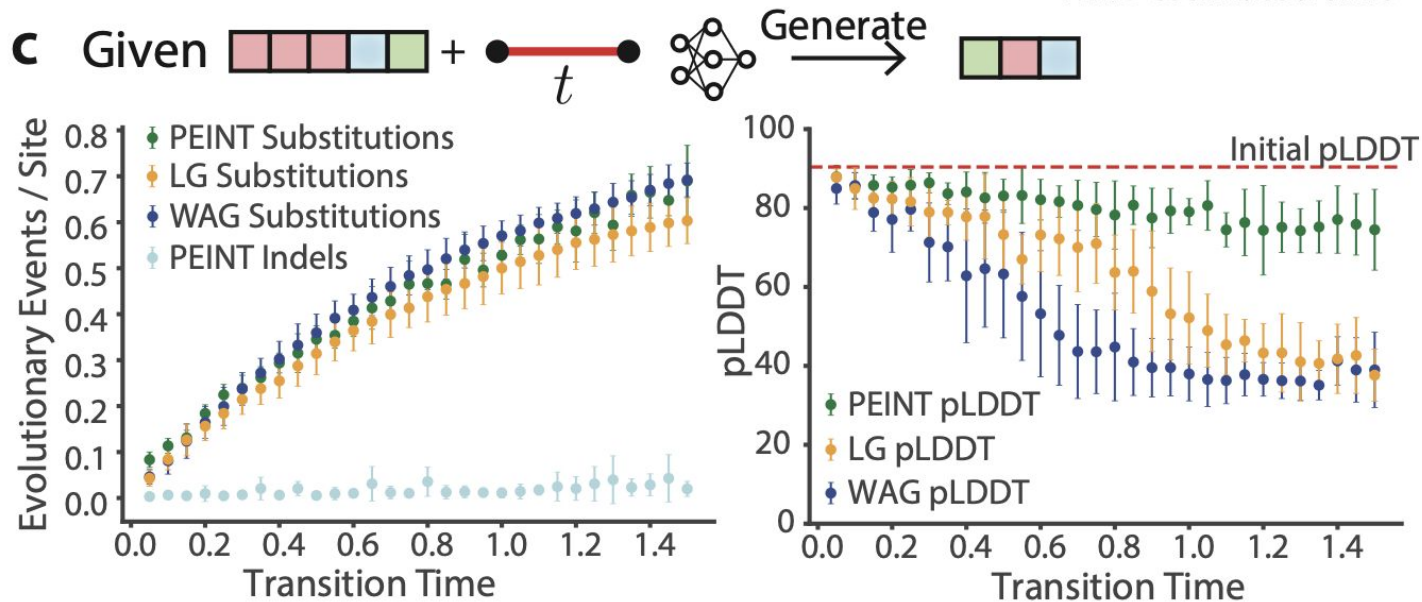
Estimate:

$$\hat{t} = \arg \max_t [\log p(y \mid x, t) + \log p(x \mid y, t)]$$

Using gradient ascent on the likelihood



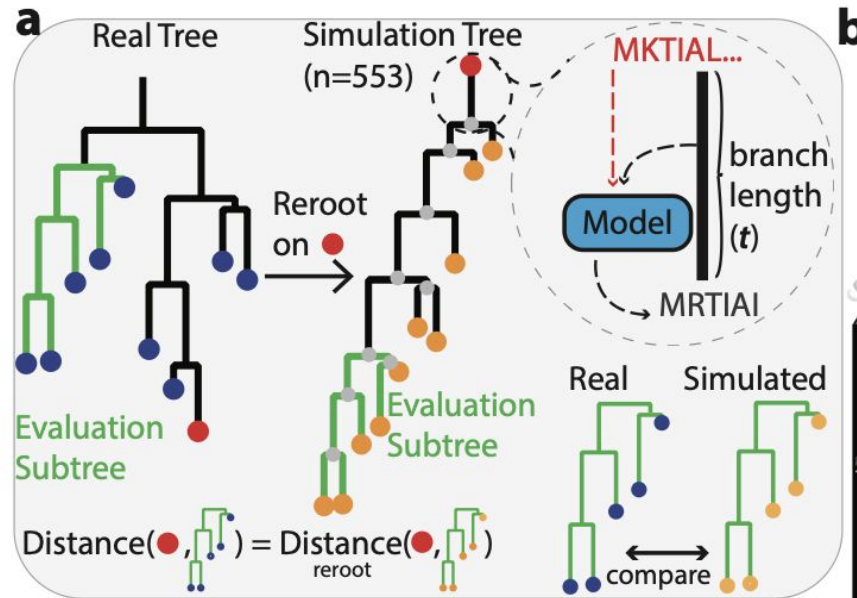
PEINT generates novel sequences with defined mutational loads while retaining high structural confidence scores



By capturing epistasis, PEINT generates compensatory mutations that preserve key functional contacts.

PEINT simulates realistic evolution within protein families along entire phylogenetic trees

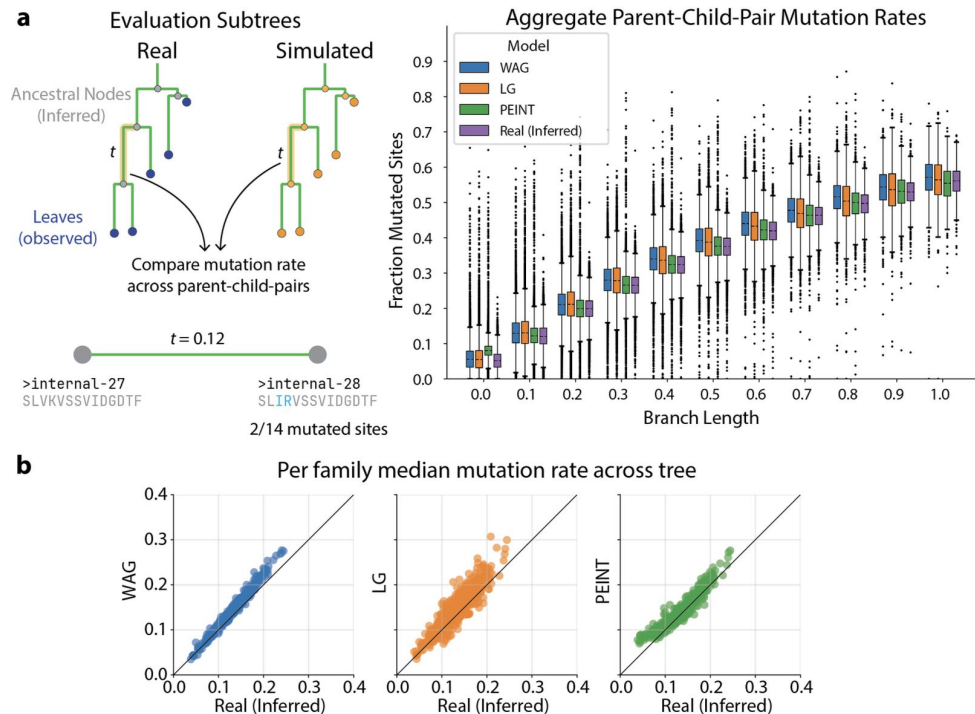
- Start from a root sequence in a held-out family.
- Recursively generate child sequences along tree branches.
- Generated child becomes the parent for the next step.
- Compare simulated leaves to real held-out leaves.



PEINT simulates realistic evolution within protein families along entire phylogenetic trees

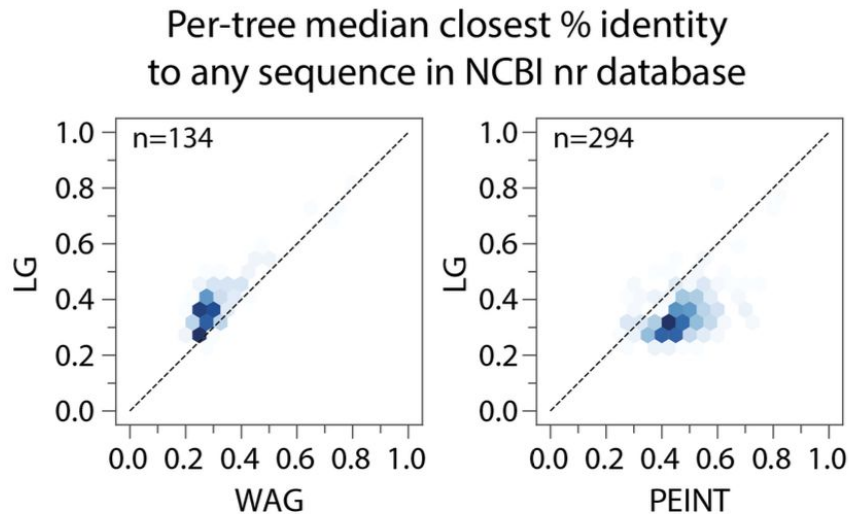
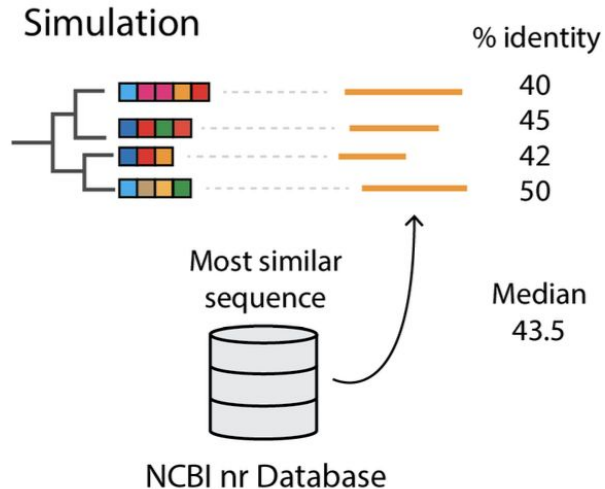
Simulations generate sequences with natural level of mutations

Mutation rates of various models across parent-child pairs were compared to **those inferred using ancestral reconstructions** from real trees and aggregated by branch length.



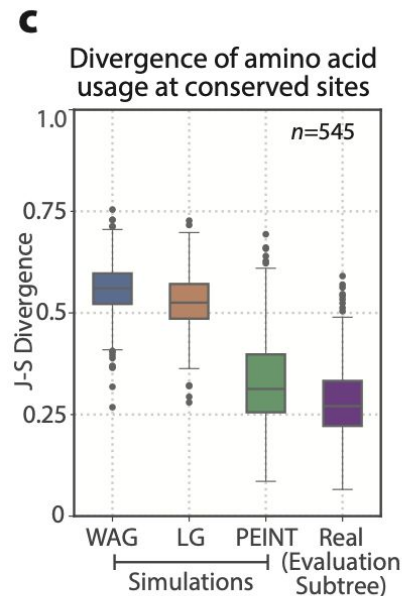
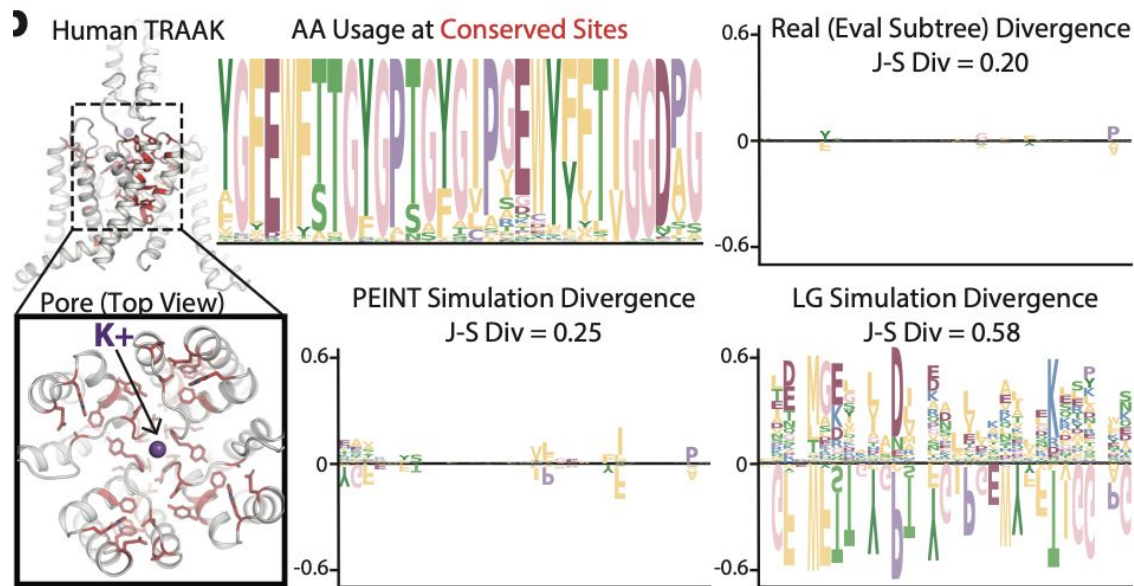
PEINT simulates realistic evolution within protein families along entire phylogenetic trees

Across all simulated trees, sequences generated by WAG, LG and PEINT shared at most 50% sequence identity with any known sequence



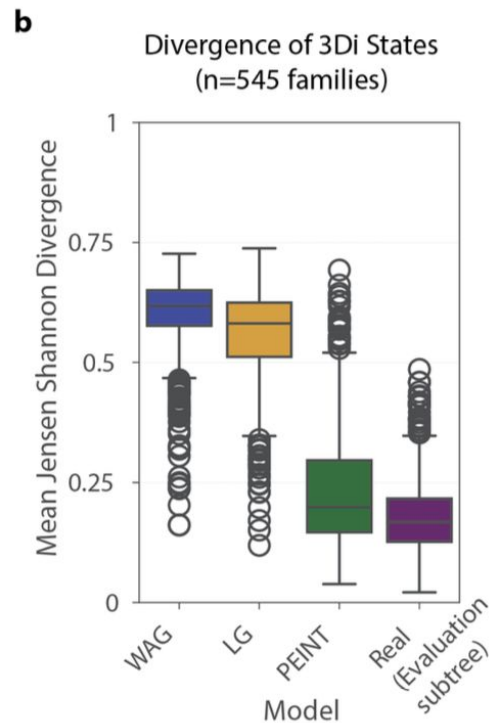
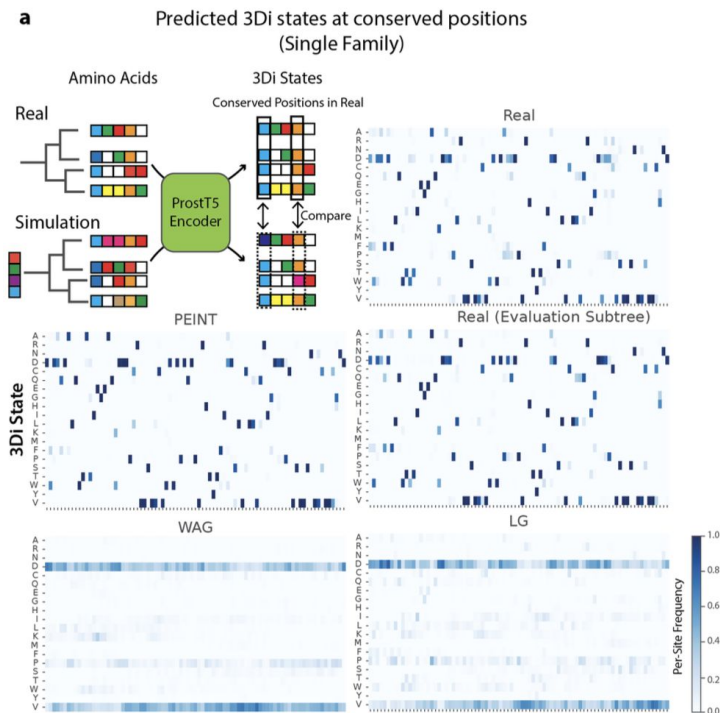
Amino acid patterns at conserved sites.

The sites that show greater than 75% conservation



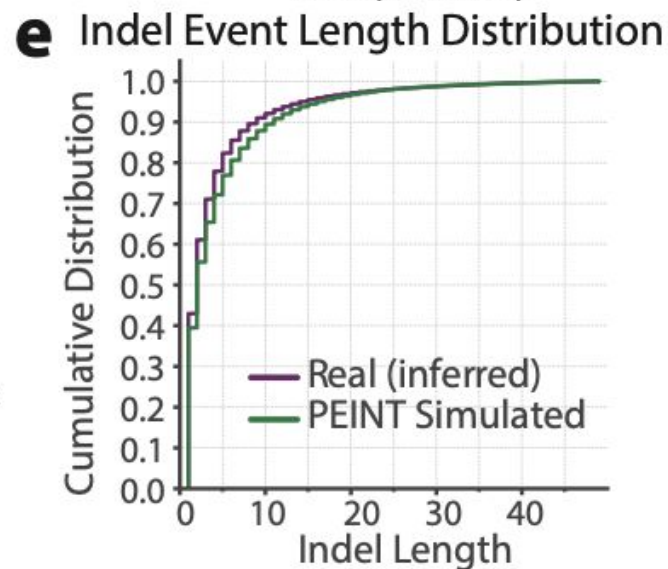
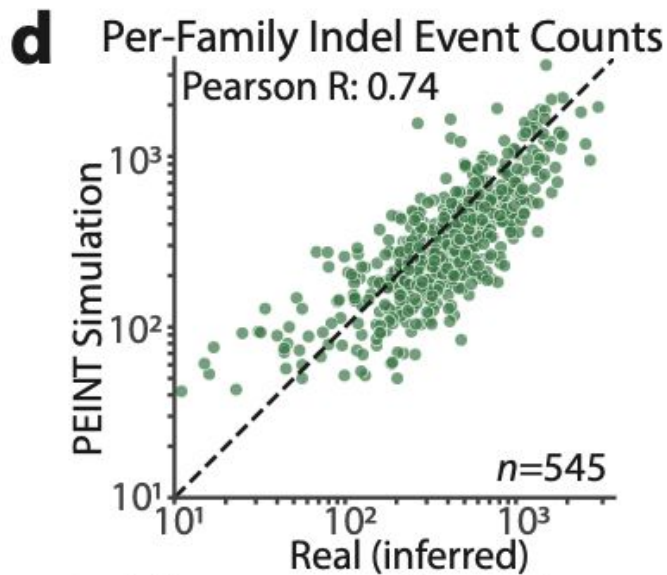
Conservation of local structural motifs

PEINT maintains conserved structural motifs that define family folds, showing divergence that is on par with natural variation



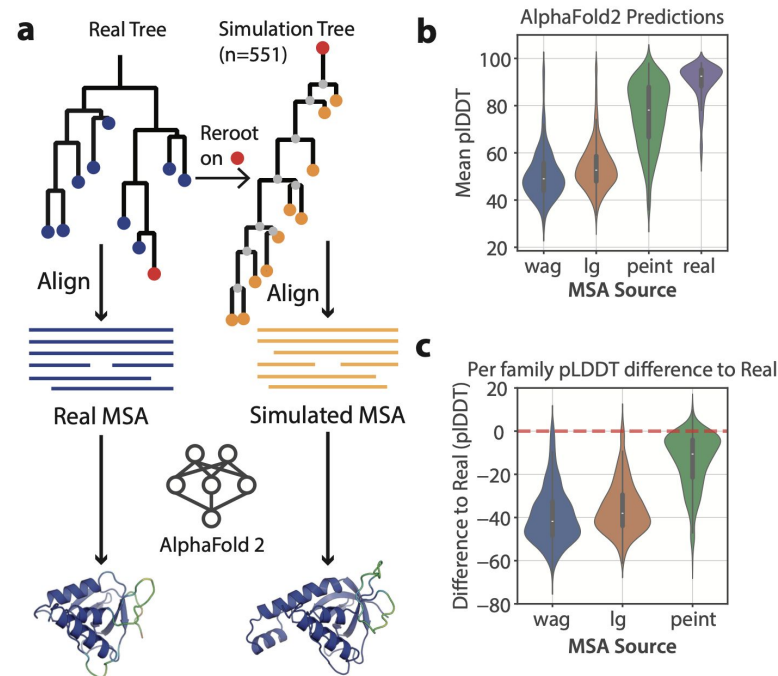
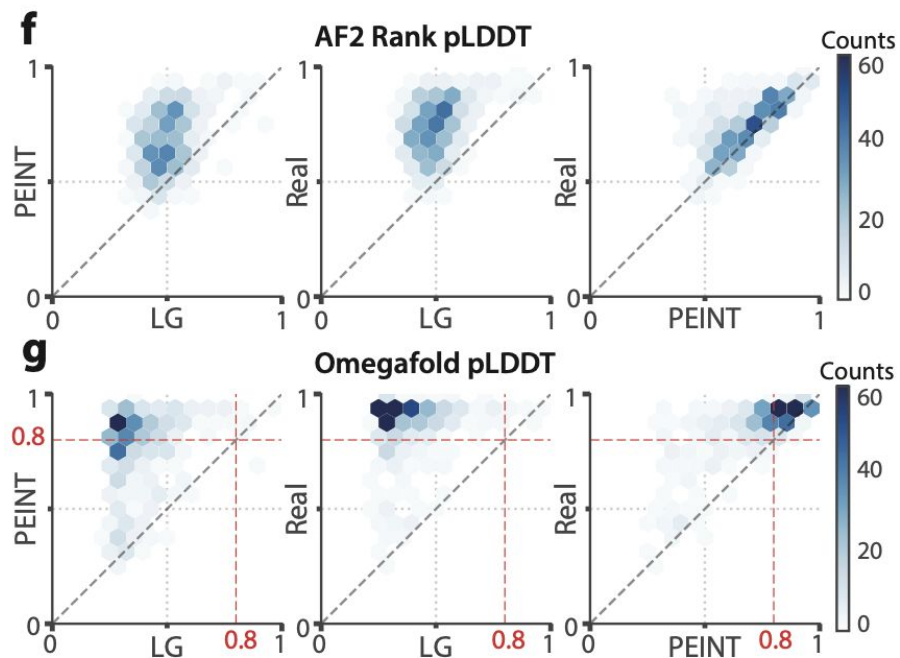
family-level insertion and deletion (indel) characteristics

PEINT faithfully captures family specific insertion/deletion (indel) rates (d) and per-event insertion length statistics

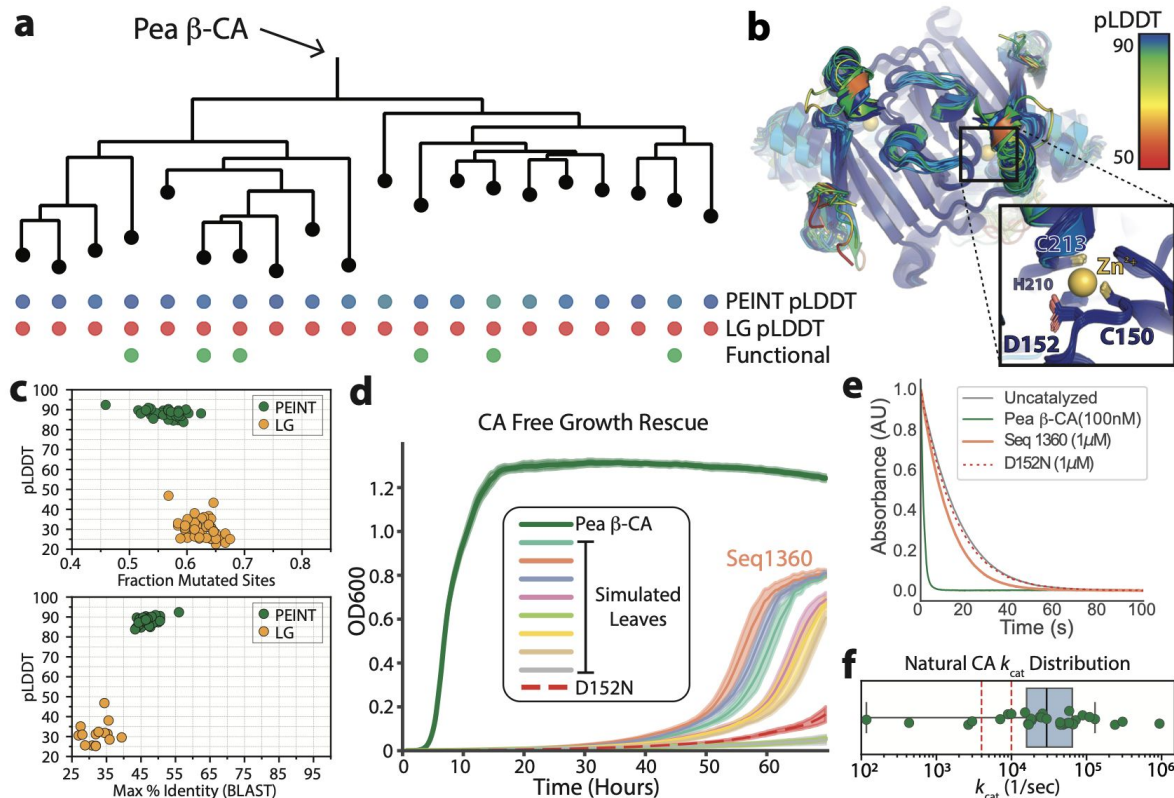


Structural confidence scores of simulated leafs

When evaluated using AF2Rank (f) or OmegaFold (g), PEINT generated leaf sequences have pLDDTs that are almost identical to real sequences, and significantly higher than those generated using the best classical model (LG).



PEINT generates functional enzymes with low sequence identity from known proteins



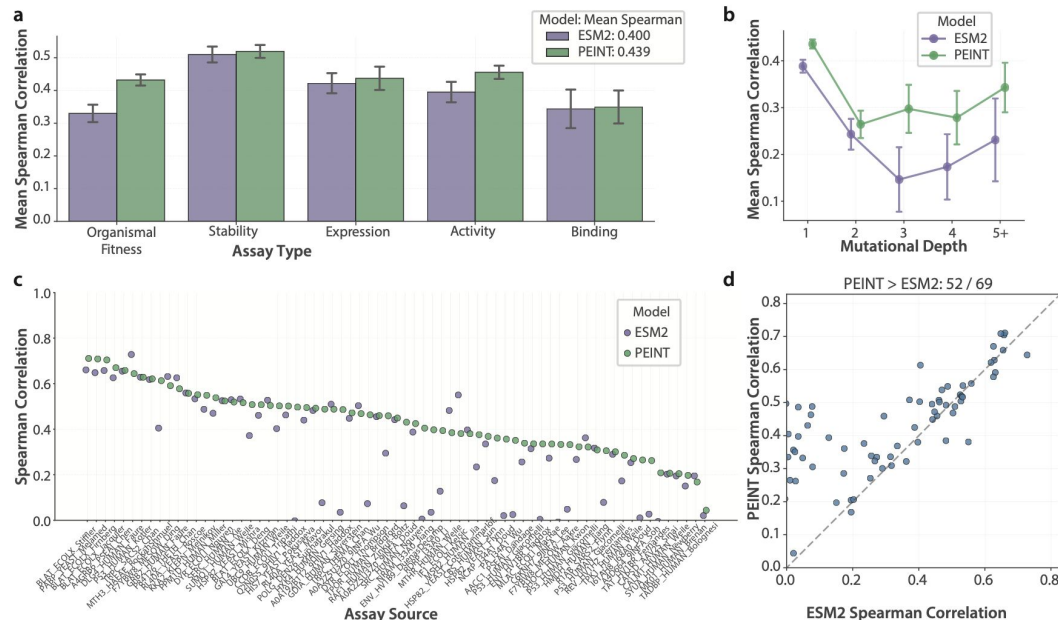
PEINT improves variant effect prediction (VEP)

a smaller PEINT with 23M on top of ESM2 (150M), using a collection of 1,000 transitions from each family in the ProteinGym.

Scoring:

$$\log \frac{p(x^{\text{mut}} | x^{\text{wt}}, t)}{p(x^{\text{wt}} | x^{\text{wt}}, t)}$$

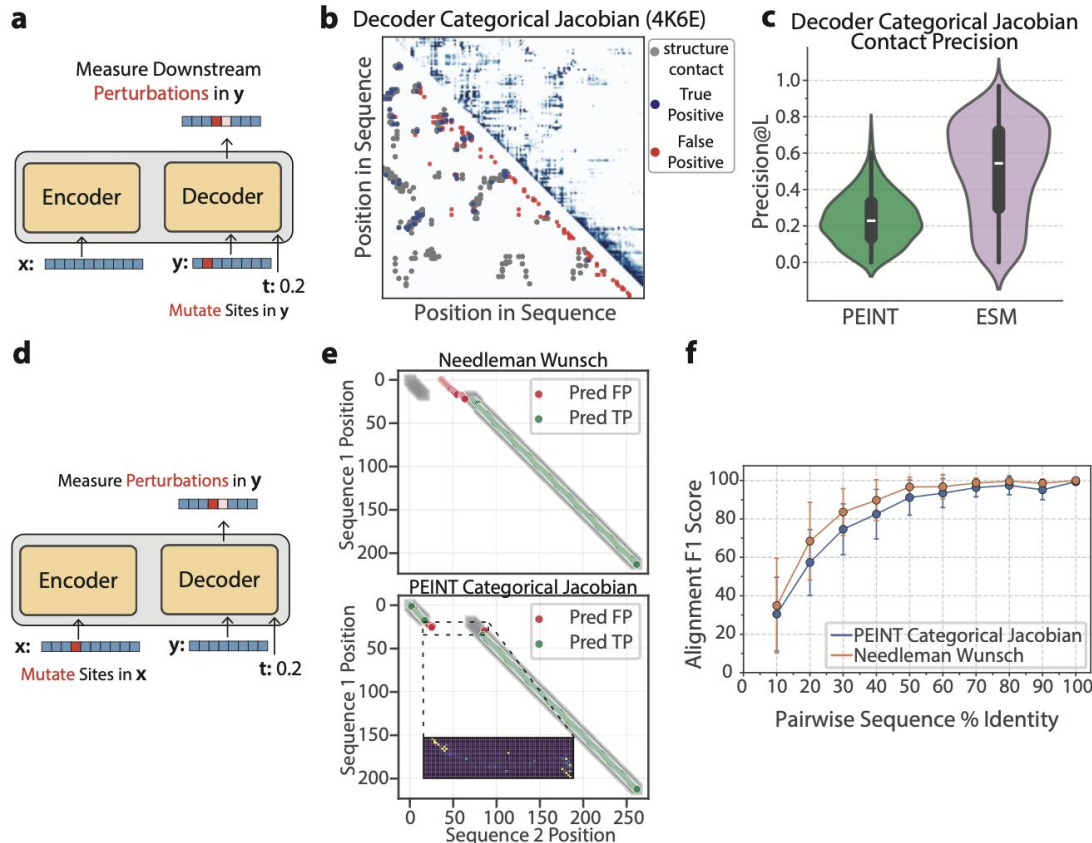
Optimal (t) varies across protein families but but changes within the training range have little effect on overall results.



Mechanistic interpretation of PEINT

Couplings learned by the PEINT decoder do correlate with structure, but more weakly than those learned by encoder-only models e.g. ESM2

PEINT encoder-decoder categorical Jacobian couplings natively recover ground truth alignments with similar accuracy as gold standard DP based alignment algorithms.



Discussion

- PEINT: A deep, alignment-free model of protein evolution in time.
- A way to use composite likelihood/cherries with transformer models.
- Better likelihood and branch-length inference than classical models.
- Realistic simulations preserving conservation, indels, and structure.
- Experimental evidence that generated divergent enzymes can function.
- Improved zero-shot variant effect prediction.

Overall: Understanding and training deep models of protein evolution on sequence data at scale, and demonstrating their effectiveness across key tasks.